Bond Yield Regression Analysis

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Figure 1: Descriptive Statistics of Predictor Variables

```
##  total_debt_ebitda ebitda_margin     market_value         sales
##  Min.   : 0.1207   Min.   : 1.538   Min.   :   6455   Min.   :  1556
##  1st Qu.: 1.8276   1st Qu.:15.880   1st Qu.:  29985   1st Qu.: 15336
##  Median : 3.1845   Median :29.257   Median :  66645   Median : 37323
##  Mean   : 4.1033   Mean   :31.672   Mean   : 296800   Mean   : 73234
##  3rd Qu.: 5.5991   3rd Qu.:44.975   3rd Qu.: 161605   3rd Qu.: 68282
##  Max.   :19.1551   Max.   :97.975   Max.   :3754848   Max.   :716924
```

```
##
##   long medium  short
##    238    666    177
```

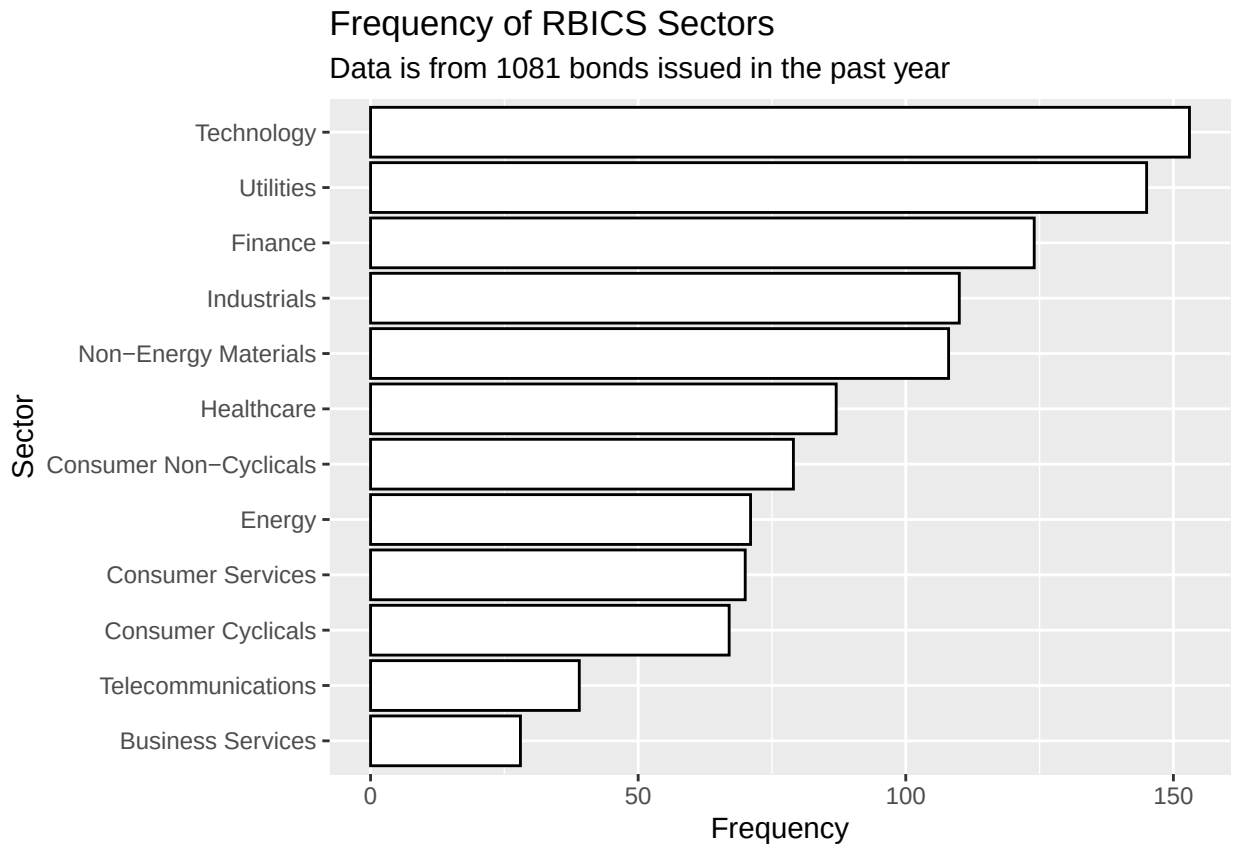
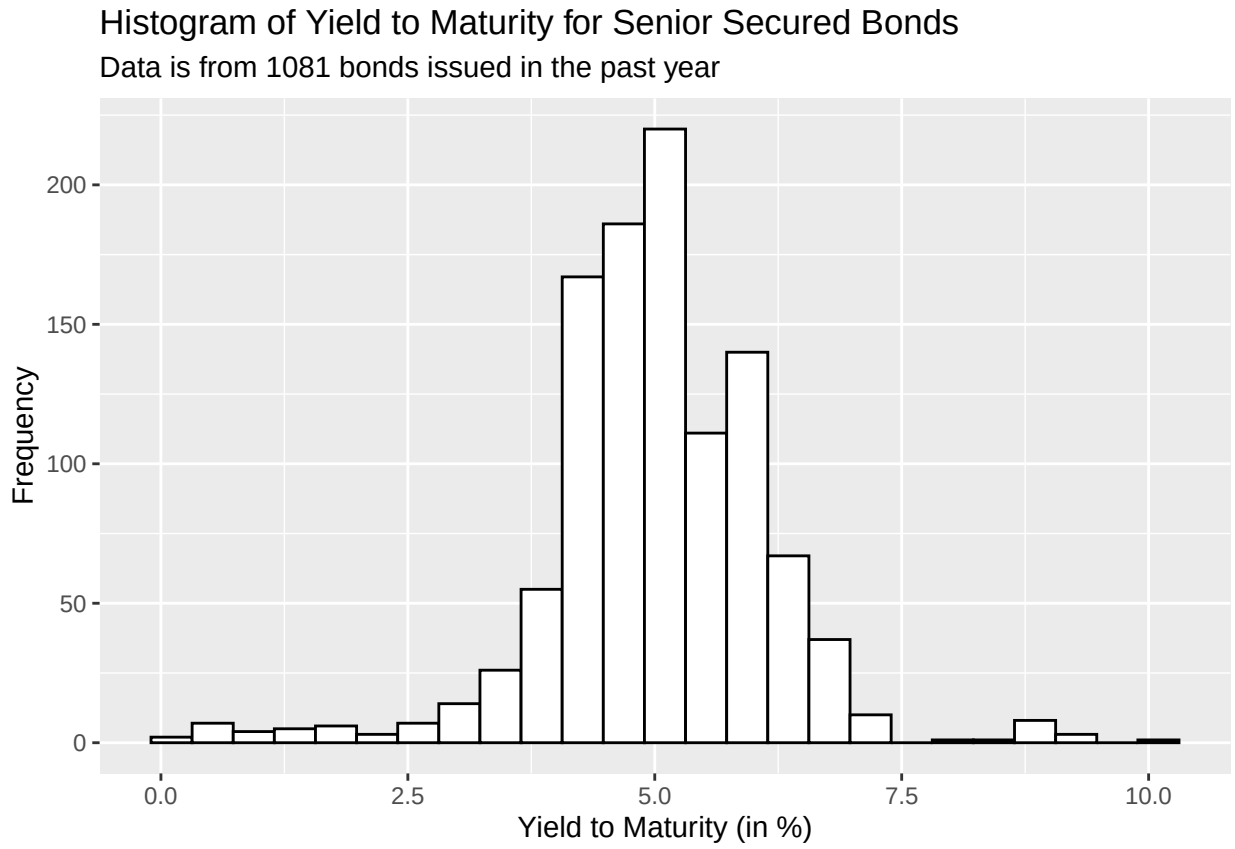


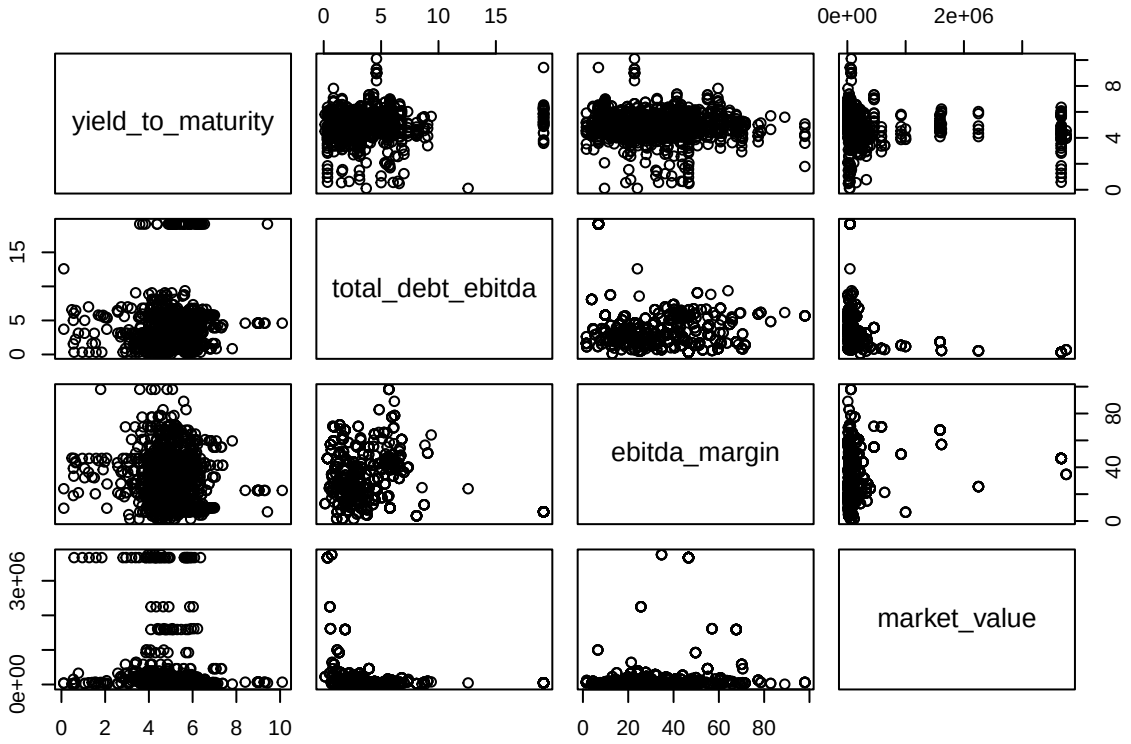
Figure 2: Descriptive Statistics of the Response Variable




```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1078  4.4342  4.9977  5.0151  5.7012 10.0993
```

```
## [1] "Standard Deviation: 1.1418"
```

Figure 3: Scatterplot Matrix and Multicollinearity Diagnostics for the Initial Model

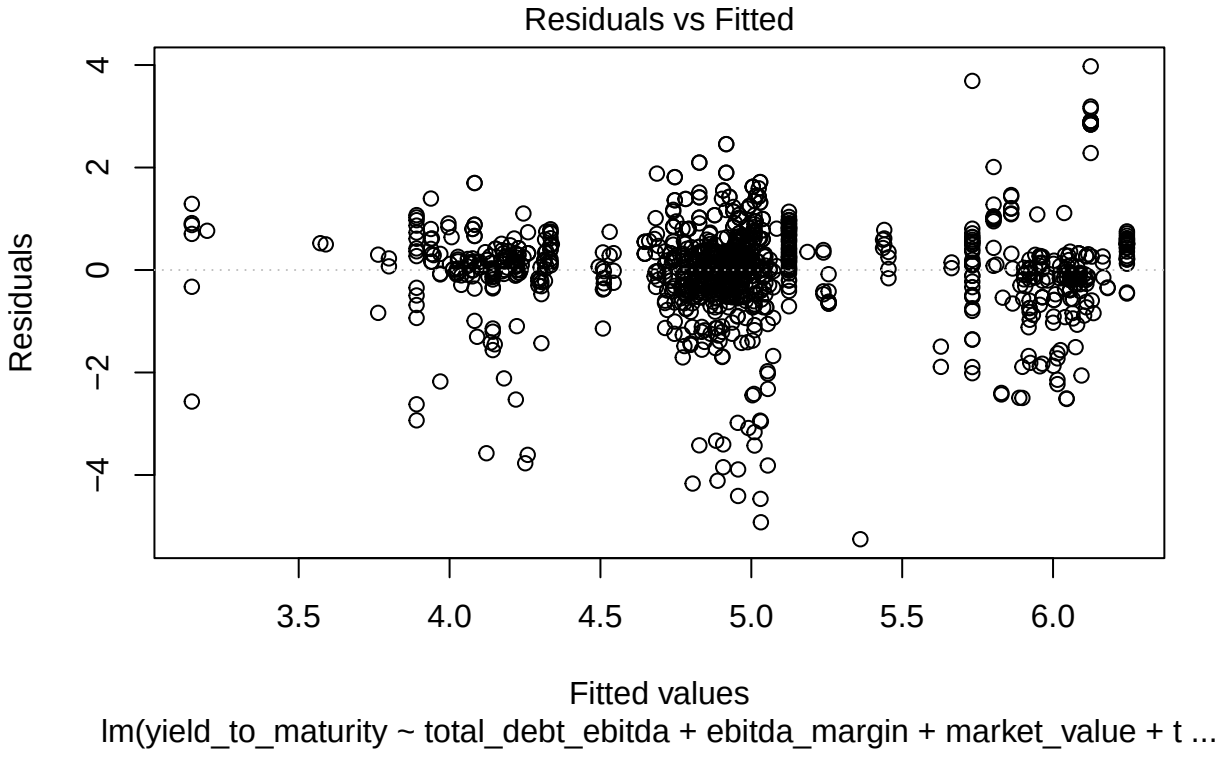



```
##                   yield_to_maturity total_debt_ebitda ebitda_margin
## yield_to_maturity         1.0000000         0.1557382    -0.1050291
## total_debt_ebitda         0.1557382         1.0000000    -0.2586028
## ebitda_margin            -0.1050291        -0.2586028     1.0000000
## market_value             -0.1612457        -0.2782214     0.2358102
##                   market_value
## yield_to_maturity   -0.1612457
## total_debt_ebitda   -0.2782214
## ebitda_margin        0.2358102
## market_value         1.0000000

## total_debt_ebitda     ebitda_margin      market_value
##          1.132307          1.106169          1.118796
```

Figure 4: Initial Model Summary and Diagnostic Plots

```
##
## Call:
## lm(formula = yield_to_maturity ~ total_debt_ebitda + ebitda_margin +
##     market_value + time_to_maturity, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2536 -0.3175  0.0749  0.4219  3.9746
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)              6.037e+00  9.166e-02  65.869  < 2e-16 ***
## total_debt_ebitda        4.473e-02  8.254e-03   5.419 7.38e-08 ***
## ebitda_margin           -4.525e-03  1.613e-03  -2.805  0.00513 **
## market_value            -2.263e-07  4.068e-08  -5.564 3.32e-08 ***
## time_to_maturitymedium -1.120e+00  7.269e-02 -15.408  < 2e-16 ***
## time_to_maturityshort  -1.865e+00  9.521e-02 -19.586  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9529 on 1075 degrees of freedom
## Multiple R-squared:  0.3067, Adjusted R-squared:  0.3035
## F-statistic: 95.13 on 5 and 1075 DF,  p-value: < 2.2e-16
```



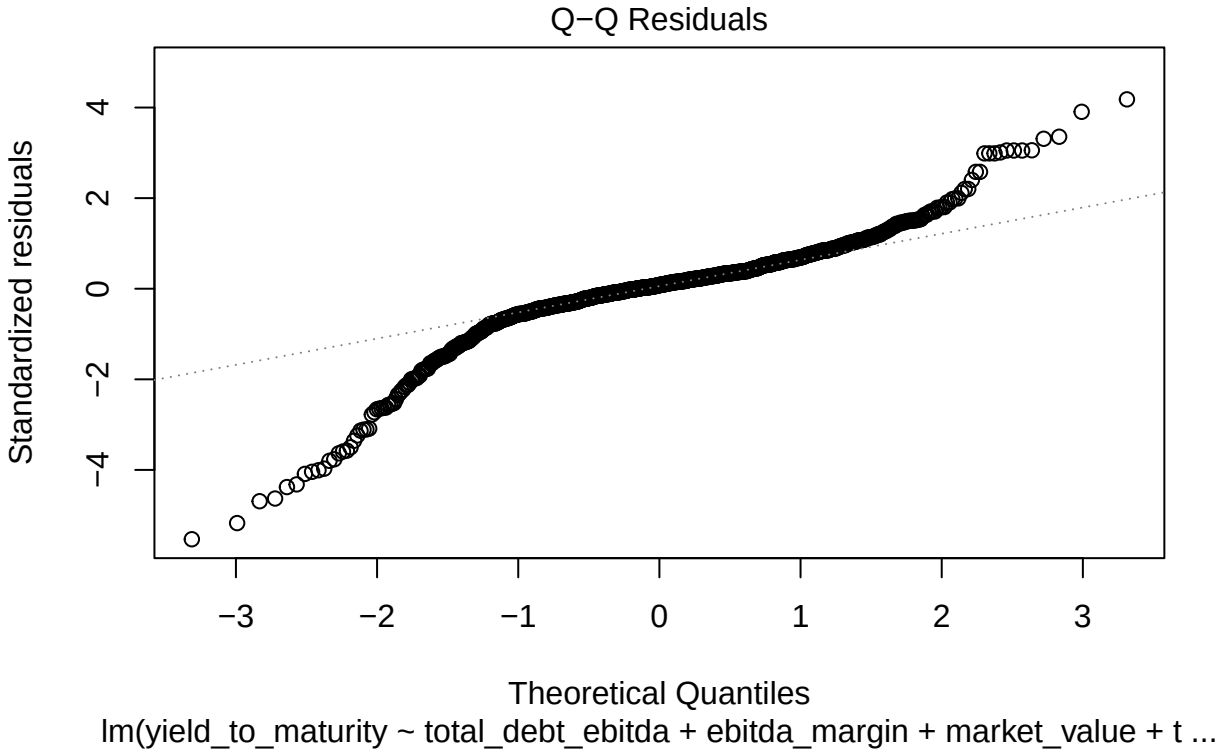

Q−Q Residuals
Standardized residuals
−4
−2
0
2
4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(yield_to_maturity ~ total_debt_ebitda + ebitda_margin + market_value + t ...

Figure 5: Log Transformation for *Yield To Maturity*

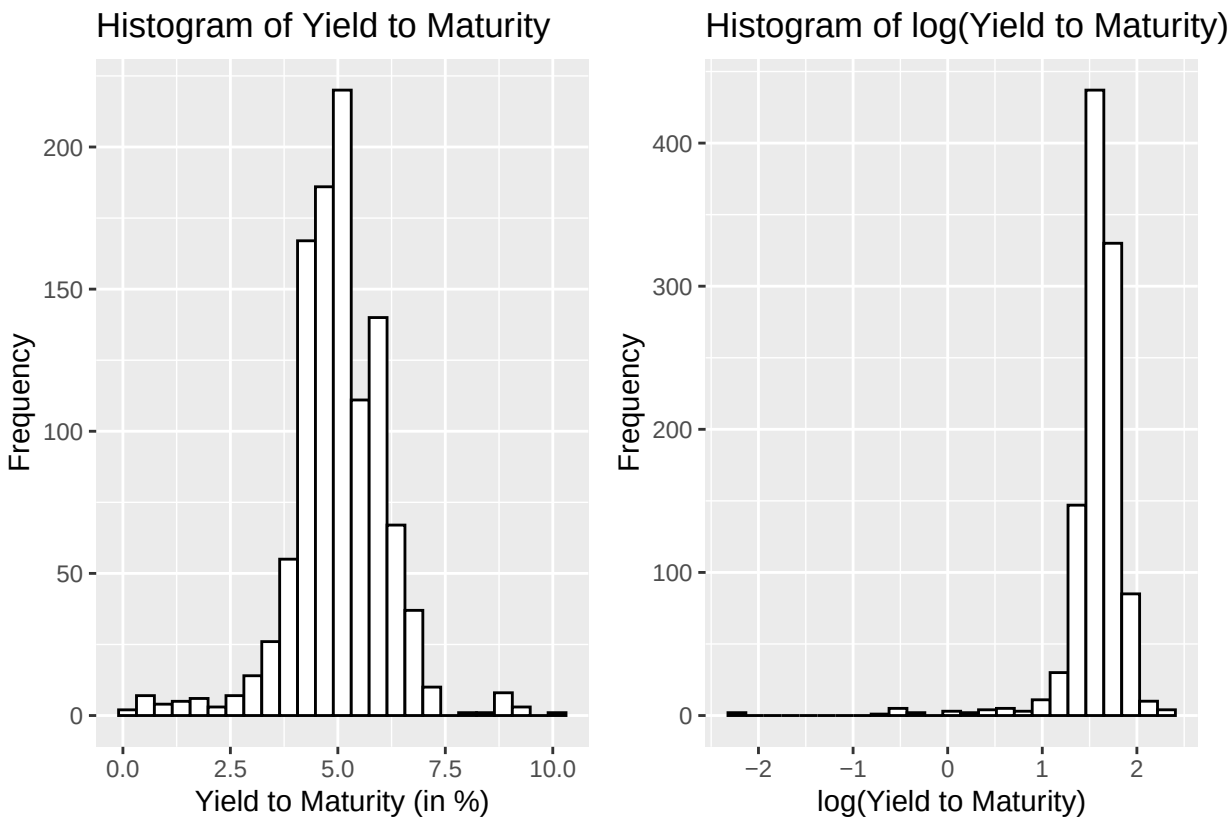


Figure 6: Log Transformation for *EBITDA Margin*

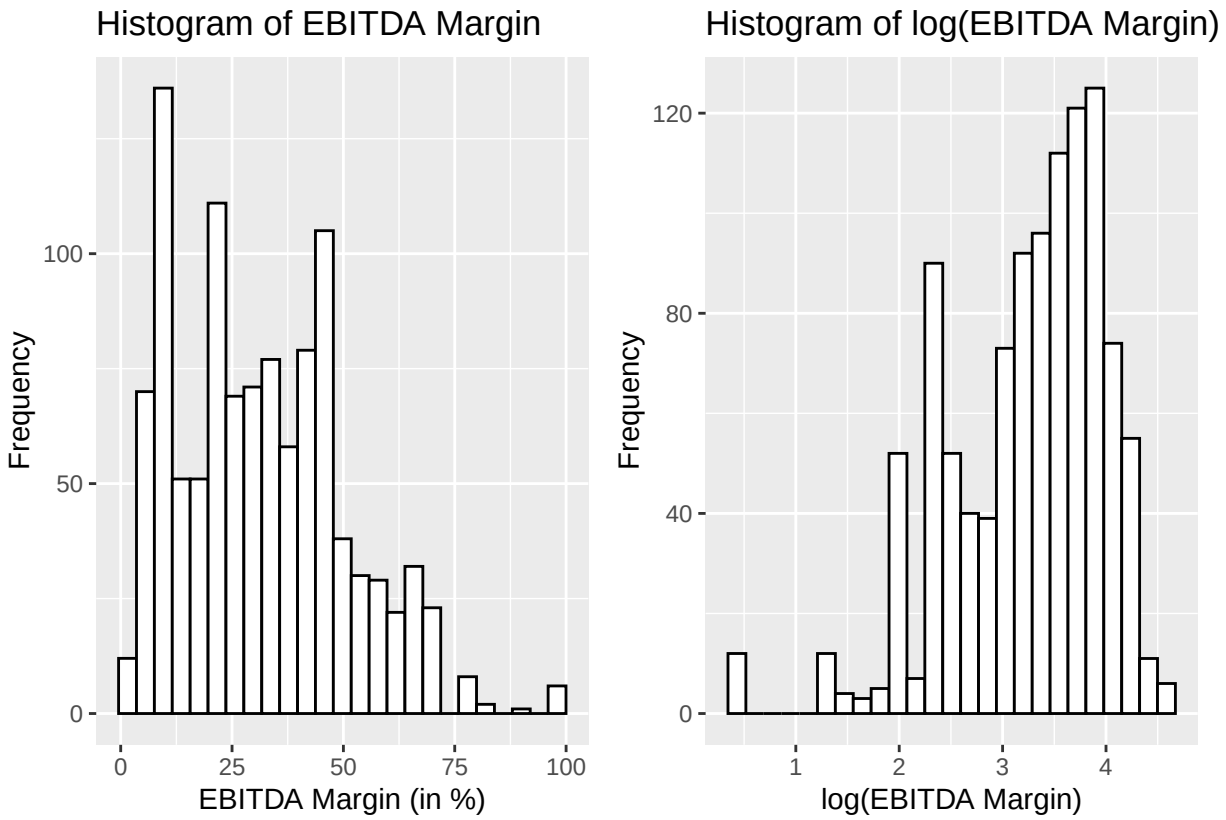


Figure 7: Log Transformation for *Market Value*

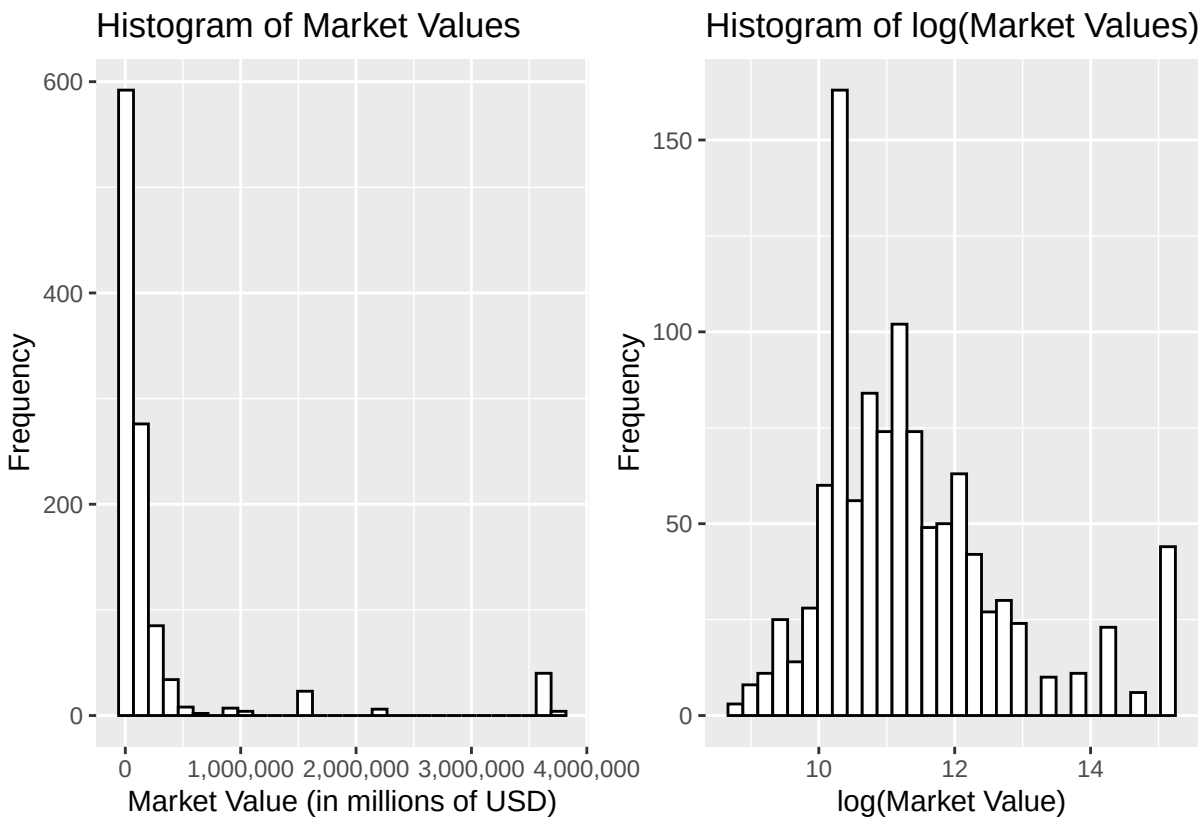


Figure 8: Log Transformation for *Debt/EBITDA Ratio*

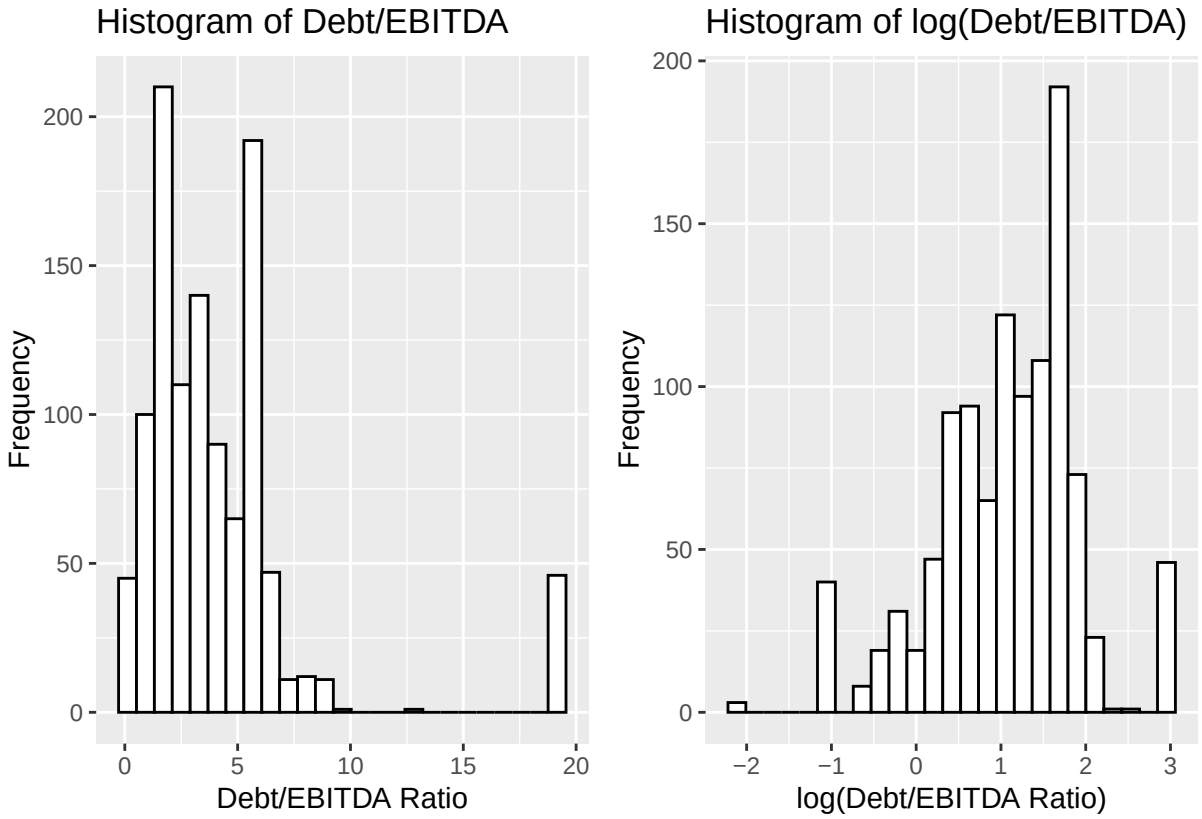
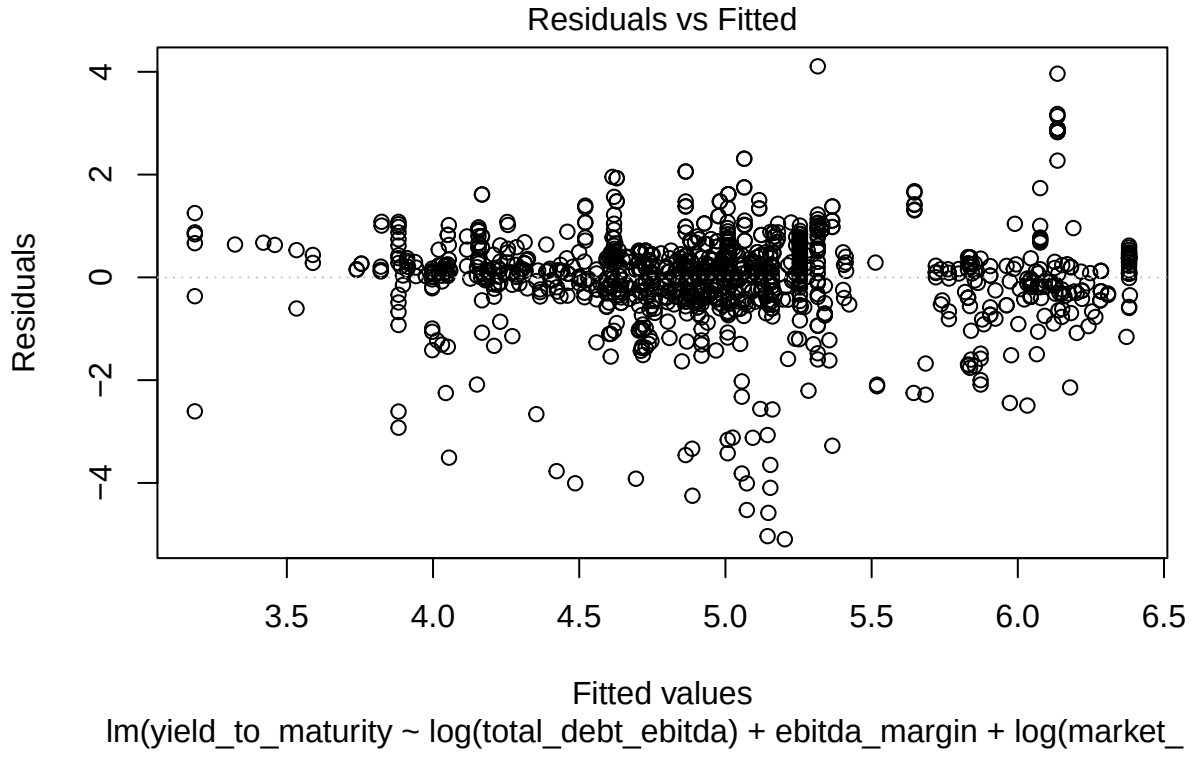


Figure 9: Model Summary and Diaganostic Plots With Log(*Debt/Ebitda Ratio*)

```
##
## Call:
## lm(formula = yield_to_maturity ~ log(total_debt_ebitda) + ebitda_margin +
##     log(market_value) + time_to_maturity, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.0954 -0.3162  0.0838  0.4231  4.1065
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)               7.993145   0.334867  23.870  < 2e-16 ***
## log(total_debt_ebitda)    0.124791   0.041130   3.034  0.00247 **
## ebitda_margin            -0.004201   0.001587  -2.647  0.00823 **
## log(market_value)        -0.175441   0.026870  -6.529 1.02e-10 ***
## time_to_maturitymedium -1.125796   0.071808 -15.678  < 2e-16 ***
## time_to_maturityshort  -1.822459   0.093965 -19.395  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9425 on 1075 degrees of freedom
## Multiple R-squared:  0.3218, Adjusted R-squared:  0.3187
## F-statistic:   102 on 5 and 1075 DF,  p-value: < 2.2e-16
```



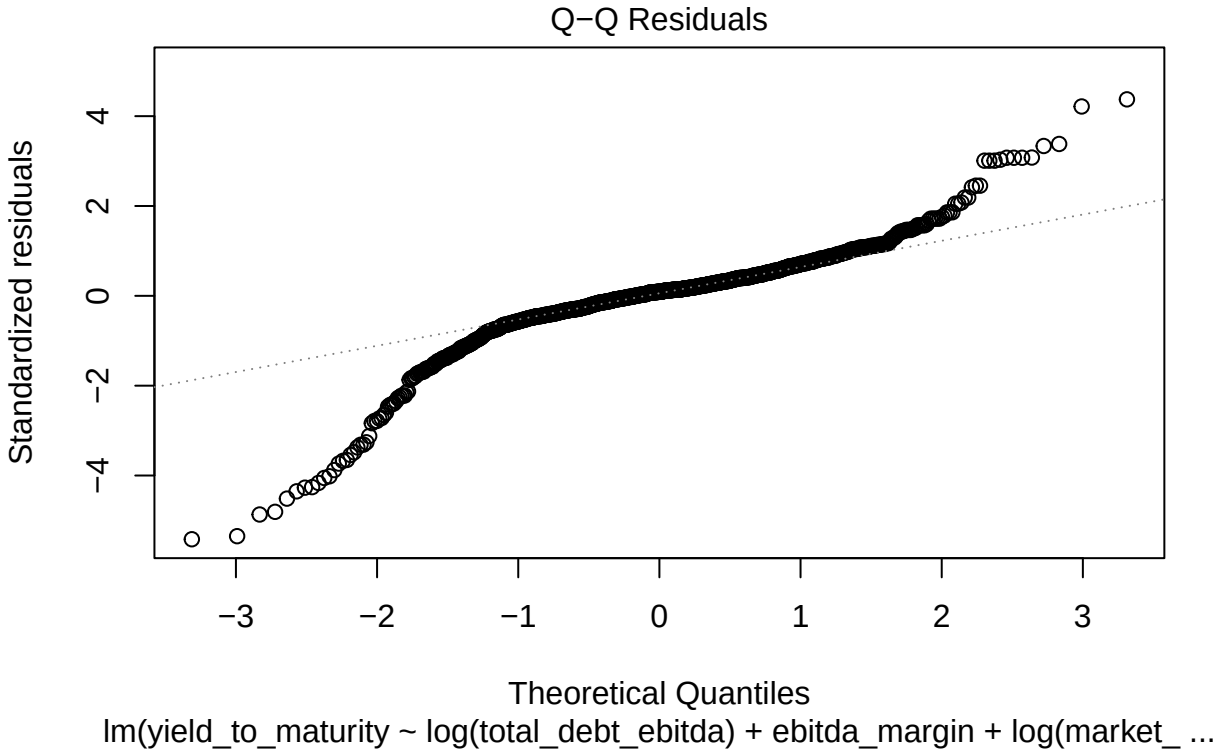
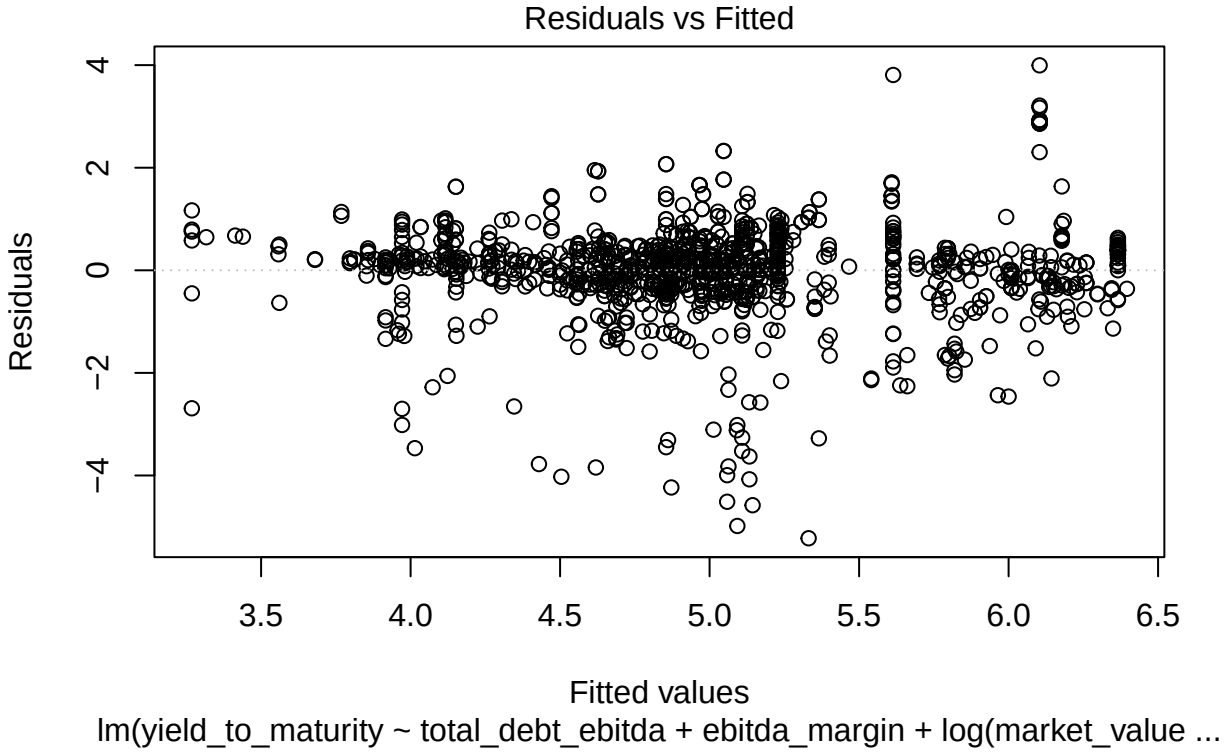

Q−Q Residuals
Standardized residuals
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(yield_to_maturity ~ log(total_debt_ebitda) + ebitda_margin + log(market_ ...

Figure 10: Model Summary and Diaganostic Plots Without Log(*Debt/Ebitda Ratio*)

```
##
## Call:
## lm(formula = yield_to_maturity ~ total_debt_ebitda + ebitda_margin +
##     log(market_value) + time_to_maturity, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2235 -0.3103  0.0993  0.4224  3.9956
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)               8.134470   0.282694  28.775  < 2e-16 ***
## total_debt_ebitda         0.036302   0.008247   4.402 1.18e-05 ***
## ebitda_margin            -0.003330   0.001600  -2.081   0.0377 *
## log(market_value)        -0.190688   0.023443  -8.134 1.14e-15 ***
## time_to_maturitymedium -1.137433   0.071565 -15.894  < 2e-16 ***
## time_to_maturityshort  -1.840540   0.093683 -19.646  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9381 on 1075 degrees of freedom
## Multiple R-squared:  0.3281, Adjusted R-squared:  0.325
## F-statistic:   105 on 5 and 1075 DF,  p-value: < 2.2e-16
```



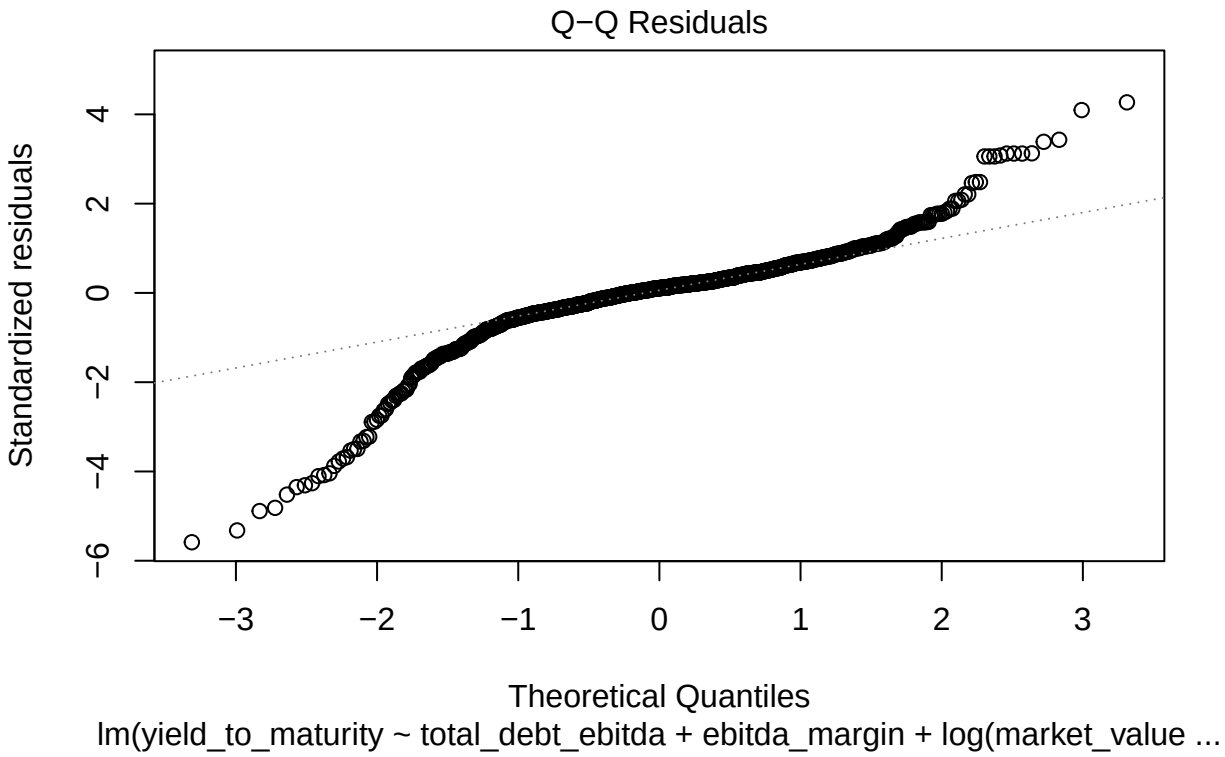

Q−Q Residuals
Standardized residuals
−6
−4
−2
0
2
4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(yield_to_maturity ~ total_debt_ebitda + ebitda_margin + log(market_value ...

Figure 11: Scatterplot Matrix After Appropriate Log Transformations

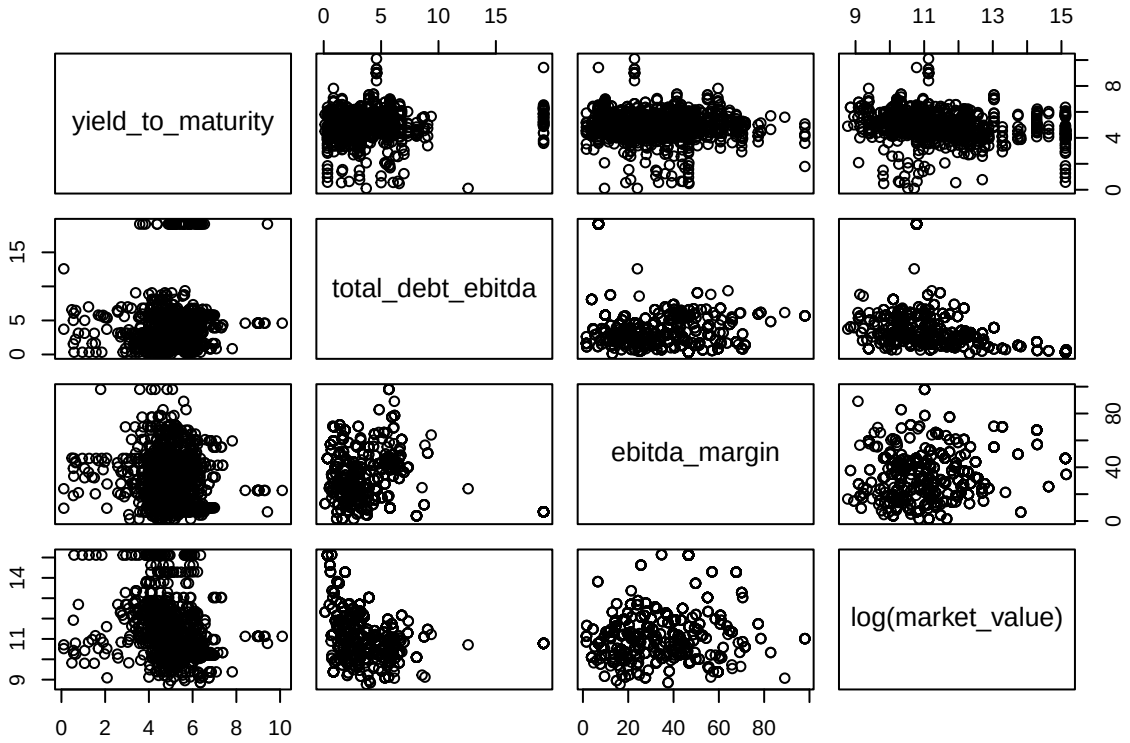


Figure 12: Multicollinearity Diagnostics After Appropriate Log Transformations

```
##                   yield_to_maturity total_debt_ebitda ebitda_margin
## yield_to_maturity         1.0000000         0.1557382    -0.1050291
## total_debt_ebitda         0.1557382         1.0000000    -0.2586028
## ebitda_margin            -0.1050291        -0.2586028     1.0000000
## log(market_value)        -0.2414843        -0.3362041     0.2775711
##                   log(market_value)
## yield_to_maturity        -0.2414843
## total_debt_ebitda        -0.3362041
## ebitda_margin             0.2775711
## log(market_value)         1.0000000

## total_debt_ebitda     ebitda_margin log(market_value)
##          1.166360          1.120882          1.179213
```

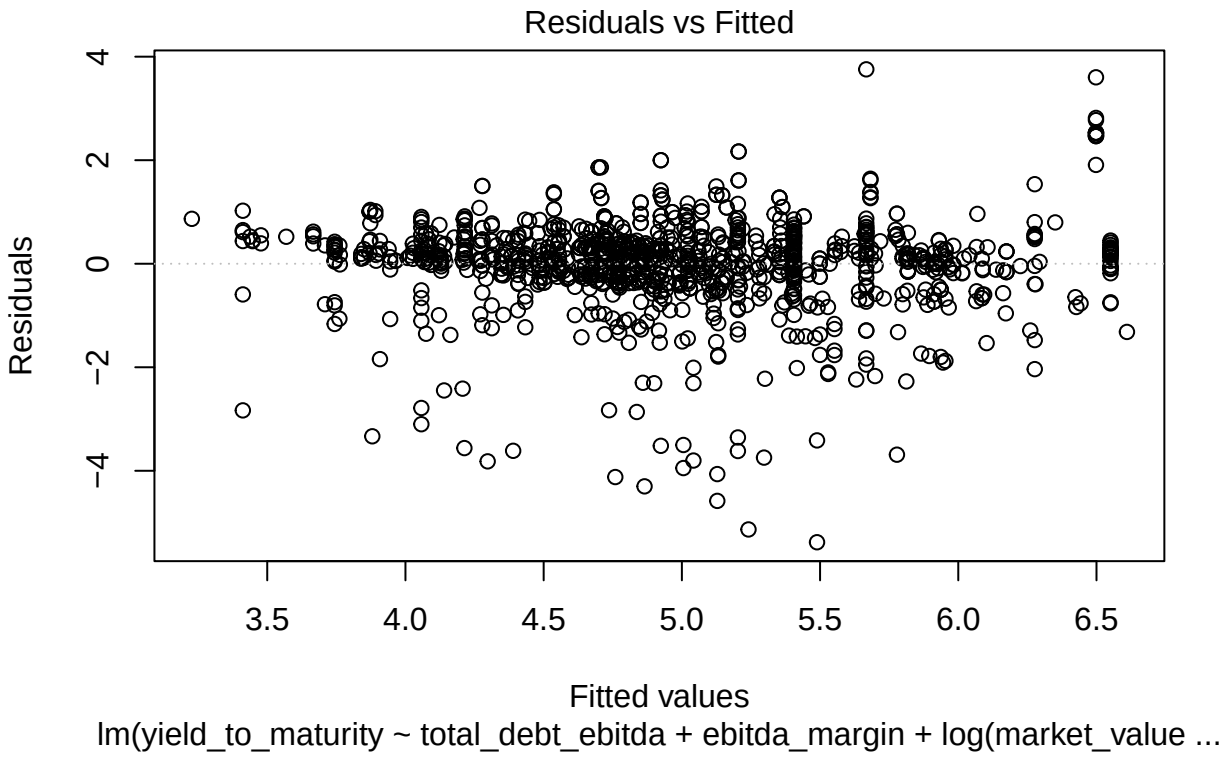
Figure 13: Model Summary with *Sales* as a Predictor

```
##
## Call:
## lm(formula = yield_to_maturity ~ total_debt_ebitda + ebitda_margin +
##     log(market_value) + time_to_maturity + sales, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2044 -0.3059  0.0984  0.4234  4.0070
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              8.272e+00  3.621e-01  22.843  < 2e-16 ***
## total_debt_ebitda        3.524e-02  8.434e-03   4.178 3.17e-05 ***
## ebitda_margin           -2.885e-03  1.761e-03  -1.638    0.102
## log(market_value)       -2.053e-01  3.358e-02  -6.112 1.38e-09 ***
## time_to_maturitymedium -1.136e+00  7.165e-02 -15.850  < 2e-16 ***
## time_to_maturityshort  -1.837e+00  9.386e-02 -19.577  < 2e-16 ***
## sales                   2.275e-07  3.753e-07   0.606    0.545
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9384 on 1074 degrees of freedom
## Multiple R-squared:  0.3284, Adjusted R-squared:  0.3246
## F-statistic: 87.51 on 6 and 1074 DF,  p-value: < 2.2e-16

## total_debt_ebitda     ebitda_margin log(market_value)             sales
##          1.217147          1.354674          2.435974          2.126905
```

Figure 14: Model Summary and Diagnostic Plots with *RBICS Economy* as a Predictor

```
##
## Call:
## lm(formula = yield_to_maturity ~ total_debt_ebitda + ebitda_margin +
##     log(market_value) + time_to_maturity + rbics_economy, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3802 -0.2660  0.1068  0.4319  3.7552
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        8.246657   0.355857  23.174  < 2e-16 ***
## total_debt_ebitda                  0.018312   0.012193   1.502   0.1334
## ebitda_margin                     -0.002097   0.002205  -0.951   0.3420
## log(market_value)                 -0.207024   0.028632  -7.231 9.17e-13 ***
## time_to_maturitymedium            -1.145260   0.072213 -15.859  < 2e-16 ***
## time_to_maturityshort             -1.791017   0.094536 -18.945  < 2e-16 ***
## rbics_economyConsumer Cyclicals    0.460991   0.258324   1.785   0.0746 .
## rbics_economyConsumer Non-Cyclicals -0.156618   0.207695  -0.754   0.4510
## rbics_economyConsumer Services     0.518677   0.207986   2.494   0.0128 *
## rbics_economyEnergy                0.081617   0.209128   0.390   0.6964
## rbics_economyFinance               0.132556   0.200376   0.662   0.5084
## rbics_economyHealthcare           -0.055511   0.203293  -0.273   0.7849
## rbics_economyIndustrials          -0.096642   0.199441  -0.485   0.6281
## rbics_economyNon-Energy Materials  0.331418   0.199809   1.659   0.0975 .
## rbics_economyTechnology            0.177084   0.202979   0.872   0.3832
## rbics_economyTelecommunications    0.249672   0.232423   1.074   0.2830
## rbics_economyUtilities            -0.147484   0.195188  -0.756   0.4501
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9209 on 1064 degrees of freedom
## Multiple R-squared:  0.3591, Adjusted R-squared:  0.3495
## F-statistic: 37.26 on 16 and 1064 DF,  p-value: < 2.2e-16
```


Residuals vs Fitted
Residuals
4
2
0
−2
−4
3.5
4.0
4.5
5.0
5.5
6.0
6.5
Fitted values
lm(yield_to_maturity ~ total_debt_ebitda + ebitda_margin + log(market_value ...

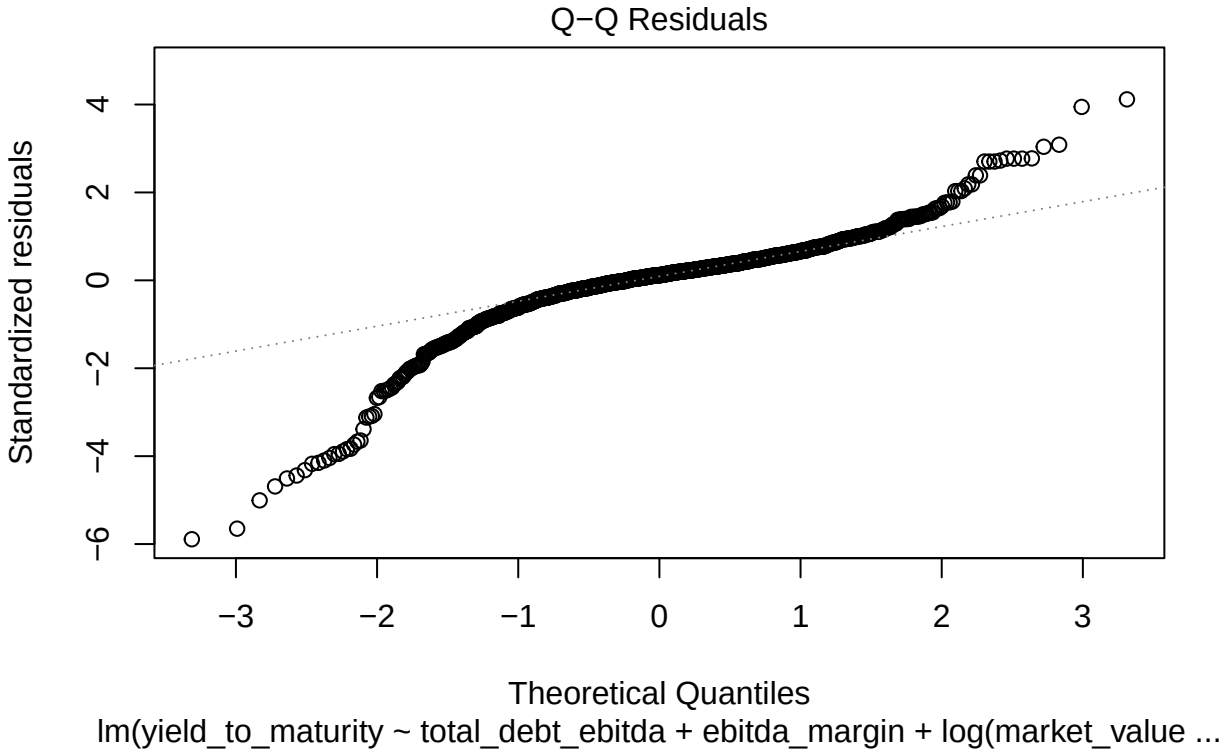

Q−Q Residuals
Standardized residuals
−6
−4
−2
0
2
4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(yield_to_maturity ~ total_debt_ebitda + ebitda_margin + log(market_value ...

Figure 15: Nested F-Test for *RBICS Economy*

```
## Analysis of Variance Table
##
## Model 1: yield_to_maturity ~ total_debt_ebitda + ebitda_margin + log(market_value) +
##     time_to_maturity
## Model 2: yield_to_maturity ~ total_debt_ebitda + time_to_maturity + ebitda_margin +
##     log(market_value) + rbics_economy
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   1075 946.02
## 2   1064 902.39 11    43.626 4.6762 5.205e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 16: Mallow's Cp for *RBICS Economy*

```
## [1] 79.5603
```

```
## [1] 112.0232
```

Figure 17: Interaction Between *Debt/EBITDA Ratio* and *Time to Maturity*

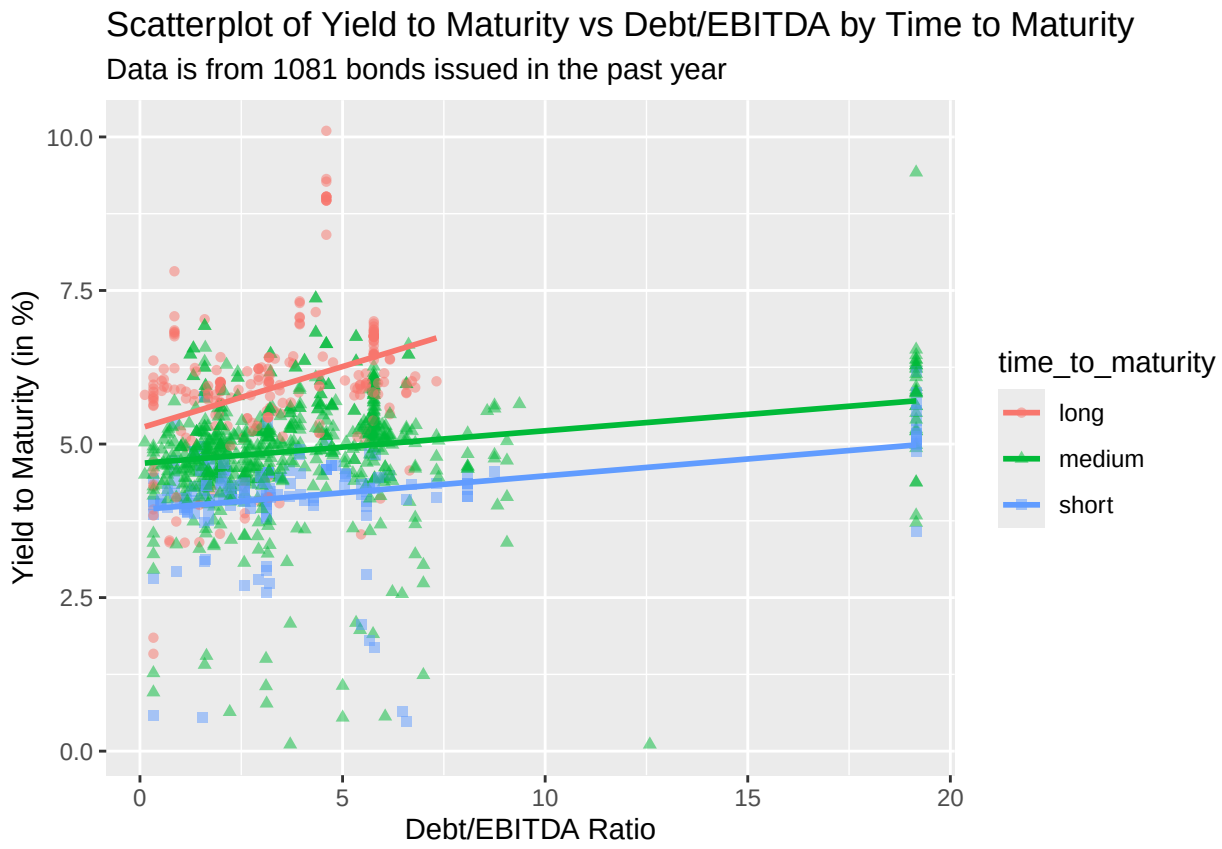
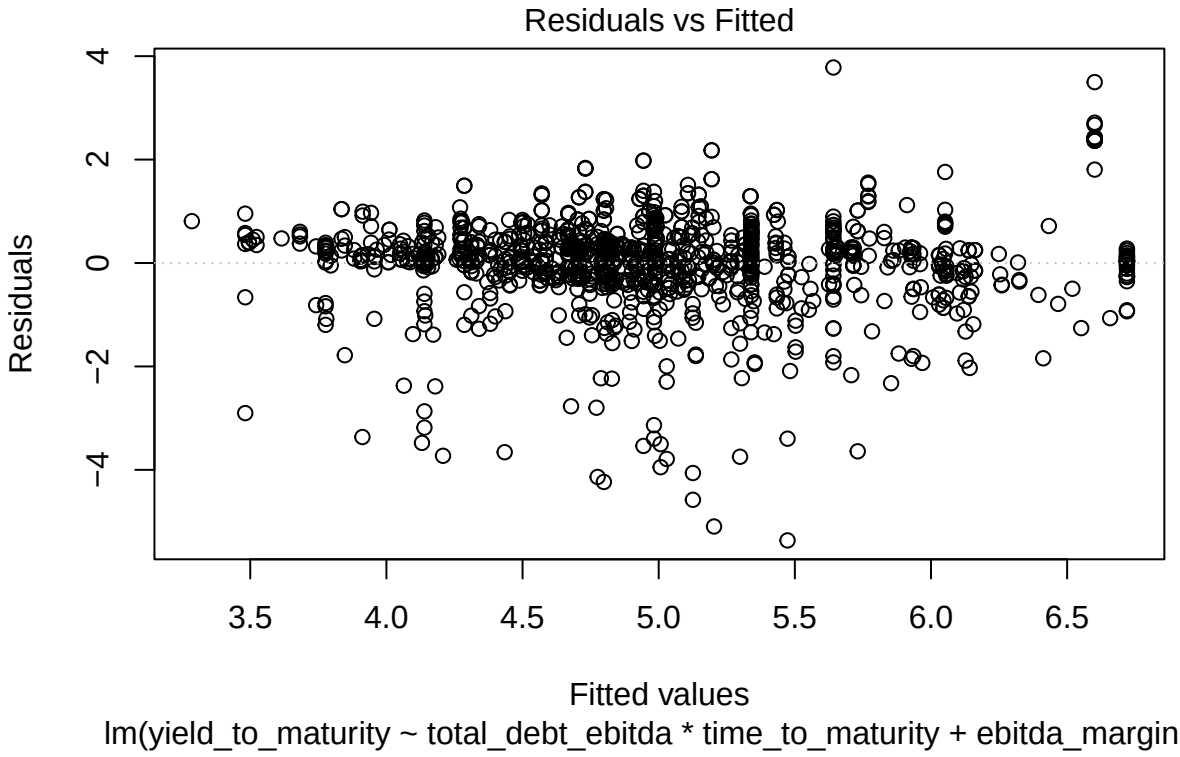


Figure 18: Model Summary and Diagnostic Plots for the *Debt/EBITDA* and *Time to Maturity* Interaction

```
##
## Call:
## lm(formula = yield_to_maturity ~ total_debt_ebitda * time_to_maturity +
##     ebitda_margin + log(market_value) + rbics_economy, data = bond_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3644 -0.2737  0.0886  0.4284  3.7808
##
## Coefficients:
##                                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                              7.811818   0.385701  20.254  < 2e-16
## total_debt_ebitda                        0.111438   0.034562   3.224  0.00130
## time_to_maturitymedium                  -0.810313   0.138033  -5.870 5.81e-09
## time_to_maturityshort                   -1.468468   0.157094  -9.348  < 2e-16
## ebitda_margin                           -0.002313   0.002200  -1.051  0.29343
## log(market_value)                       -0.194161   0.028912  -6.716 3.05e-11
## rbics_economyConsumer Cyclicals          0.504506   0.258230   1.954  0.05100
## rbics_economyConsumer Non-Cyclicals     -0.165812   0.207151  -0.800  0.42363
## rbics_economyConsumer Services           0.490028   0.207627   2.360  0.01845
## rbics_economyEnergy                      0.104830   0.208674   0.502  0.61552
## rbics_economyFinance                     0.127335   0.199819   0.637  0.52410
## rbics_economyHealthcare                 -0.056011   0.202707  -0.276  0.78236
## rbics_economyIndustrials                -0.093457   0.198872  -0.470  0.63850
## rbics_economyNon-Energy Materials        0.268418   0.200445   1.339  0.18082
## rbics_economyTechnology                  0.176547   0.202417   0.872  0.38330
## rbics_economyTelecommunications          0.238793   0.231790   1.030  0.30314
## rbics_economyUtilities                  -0.205387   0.195663  -1.050  0.29410
## total_debt_ebitda:time_to_maturitymedium -0.098681   0.034502  -2.860  0.00432
## total_debt_ebitda:time_to_maturityshort  -0.098516   0.036745  -2.681  0.00745
##
## (Intercept)                             ***
## total_debt_ebitda                       **
## time_to_maturitymedium                  ***
## time_to_maturityshort                   ***
## ebitda_margin
## log(market_value)                       ***
## rbics_economyConsumer Cyclicals         .
## rbics_economyConsumer Non-Cyclicals
## rbics_economyConsumer Services          *
## rbics_economyEnergy
## rbics_economyFinance
## rbics_economyHealthcare
## rbics_economyIndustrials
## rbics_economyNon-Energy Materials
## rbics_economyTechnology
## rbics_economyTelecommunications
## rbics_economyUtilities
## total_debt_ebitda:time_to_maturitymedium **
## total_debt_ebitda:time_to_maturityshort  **
## ---
```

```
## Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1
##
## Residual standard error: 0.9182 on 1062 degrees of freedom
## Multiple R-squared:  0.3641, Adjusted R-squared:  0.3533
## F-statistic: 33.78 on 18 and 1062 DF,  p-value: < 2.2e-16
```



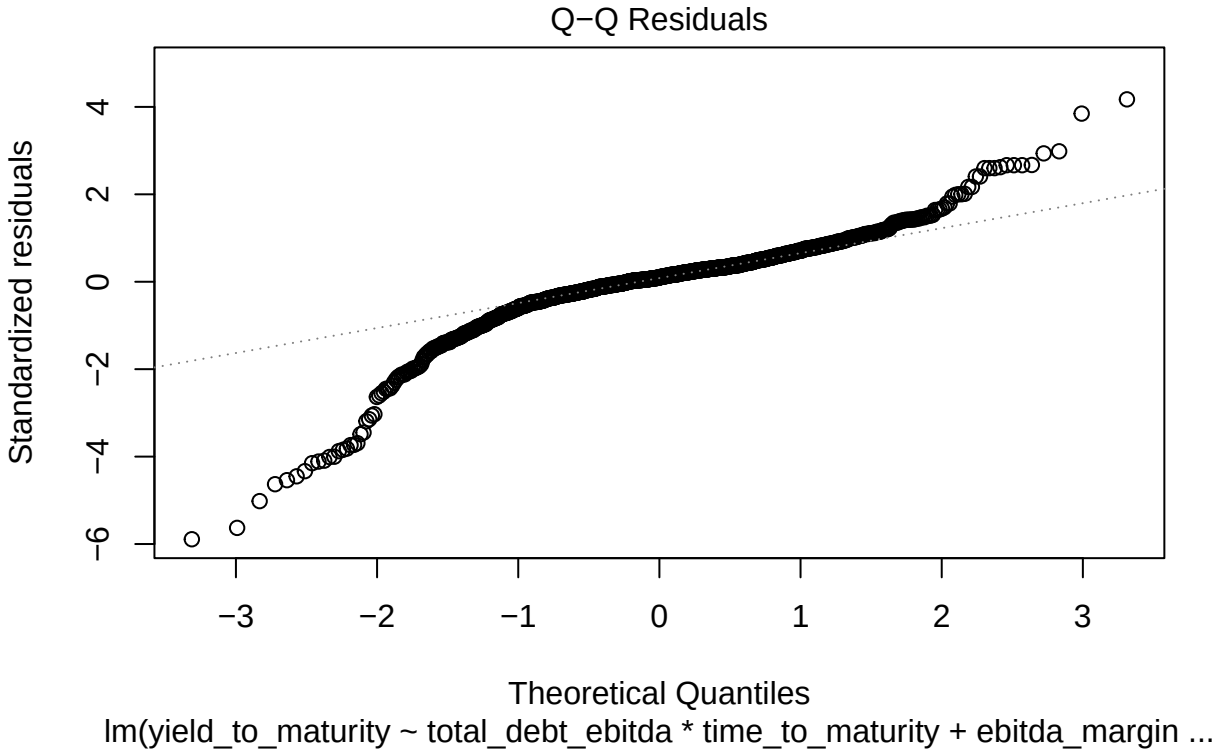

Q−Q Residuals
Standardized residuals
−6
−4
−2
0
2
4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(yield_to_maturity ~ total_debt_ebitda * time_to_maturity + ebitda_margin ...

Figure 19: Nested F-test for the *Debt/EBITDA* and *Time to Maturity* Interaction

```
## Analysis of Variance Table
##
## Model 1: yield_to_maturity ~ total_debt_ebitda + time_to_maturity + ebitda_margin +
##     log(market_value) + rbics_economy
## Model 2: yield_to_maturity ~ total_debt_ebitda * time_to_maturity + ebitda_margin +
##     log(market_value) + rbics_economy
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1   1064 902.39
## 2   1062 895.41  2    6.9873 4.1437 0.01612 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 20: Mallow's Cp for the *Debt/EBITDA* and *Time to Maturity* Interaction

```
## [1] 79.5603
```

```
## [1] 74.83725
```

Figure 21: Residual vs Fitted Values by *RBICS Economy*

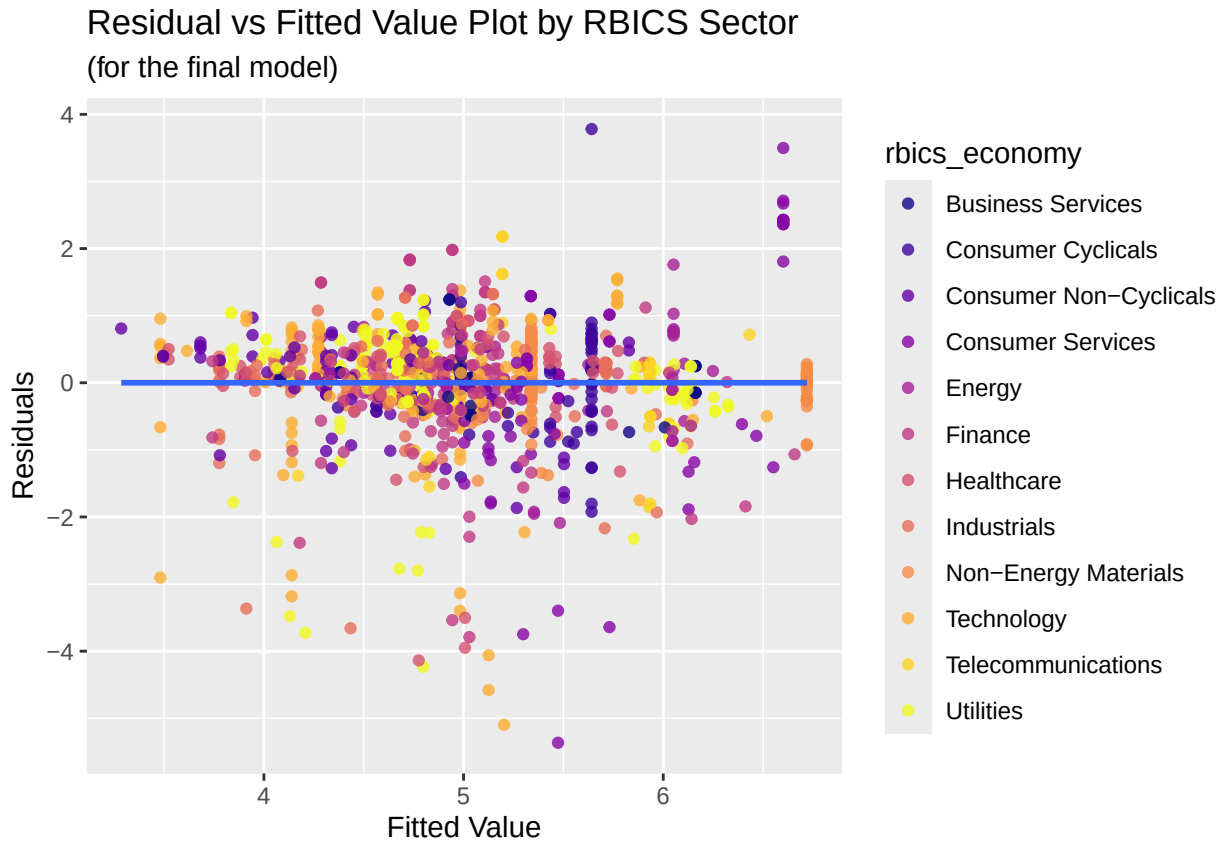


Figure 22: Residual vs Fitted Values by *Time to Maturity*

